

On the Impact of Data Splitting Strategies in Learning-Based Entity Resolution

2026-02-08

Abstract

Learning-based Entity Resolution (ER) systems rely on supervised pairwise classifiers to induce clusters of records referring to the same real-world entity. Despite advances in modeling, evaluation practices remain sensitive to how data are split into training and testing sets. In particular, random record-level splitting can introduce transductive leakage, producing optimistic estimates that do not reflect generalization to unseen entities. We formalize an entity-disjoint evaluation protocol and empirically examine its effects on two benchmark datasets. Using clustering-level and pairwise metrics, we show that splitting strategy can dominate observed performance differences and can materially alter metric conclusions.

1 Introduction

Entity Resolution (ER) aims to identify and cluster records that refer to the same real-world entity. Many modern ER pipelines adopt learning-based approaches in which a classifier is trained to predict whether pairs of records match, and a clustering procedure is applied to the resulting similarity graph.

A central difficulty is that ER data violate i.i.d. assumptions: a single record appears in many candidate pairs, and entity-level structure induces strong dependencies among samples. Consequently, naive splitting of records (or candidate pairs) can create transductive leakage, allowing entity-specific information observed during training to reappear during testing. This inflates evaluation metrics in ways that may not reflect realistic deployment settings.

This paper focuses on evaluation rather than proposing a new ER model. We formalize an entity-disjoint evaluation protocol and compare it with record-level splitting. The results show that splitting strategy can systematically change conclusions drawn from commonly used metrics, particularly when pairwise and clustering-level metrics disagree.

2 Related Work

Entity Resolution (ER), also known as record linkage or duplicate detection, has been studied extensively across database, information retrieval, and statistical communities. Early surveys established ER as a problem of identifying and clustering records that refer to the same real-world entity, highlighting the challenges posed by noisy attributes, missing values, and non-trivial similarity functions (2; 3).

Learning-based ER and system frameworks. Modern ER systems increasingly rely on supervised or semi-supervised learning to perform pairwise matching, followed by clustering to enforce transitivity. Systems such as Magellan (4) demonstrate the effectiveness of learning-based matching, while also emphasizing that high pairwise accuracy does not necessarily translate into correct entity reconstruction. This distinction motivates the use of clustering-level evaluation metrics alongside pairwise measures.

Evaluation challenges and data dependence. A recurring theme in ER research is that evaluation is complicated by strong dependencies among records and candidate pairs. Statistical perspectives on ER explicitly note that naive evaluation assumptions can be misleading when data are not independent (5). Recent work has further emphasized that design choices in blocking and evaluation can substantially influence observed performance (7).

Clustering metrics and their behavior. The choice of evaluation metric plays a critical role in interpreting ER results. Amigó et al. (6) provide a systematic analysis of clustering evaluation measures, showing that metrics such as ARI, NMI, and Variation of Information (VI) respond differently to over-merging and over-splitting. Their analysis supports the view that information-theoretic metrics may improve under aggressive merging even when entity reconstruction quality deteriorates.

Unified views of ER. Recent work has sought to unify disparate ER approaches under a common conceptual framework. Binette and Steorts (8) present a modern overview of ER that emphasizes structural dependencies and evaluation complexity, reinforcing the need for principled evaluation protocols.

In contrast to prior work that focuses on developing new ER models, this paper centers on evaluation methodology. We empirically demonstrate that data splitting strategy alone can dominate observed performance differences, and we show how metric behavior under different protocols can lead to divergent—and potentially misleading—conclusions.

3 Problem Setting and Evaluation Framework

We consider a standard learning-based entity resolution (ER) setting in which records referring to the same real-world entity must be identified and grouped. Let $\mathcal{R} = \{r_1, \dots, r_n\}$ denote a collection of records and $\mathcal{E} = \{e_1, \dots, e_m\}$ the set of true entities. Each record $r_i \in \mathcal{R}$ is associated with exactly one entity via a mapping $\pi : \mathcal{R} \rightarrow \mathcal{E}$.

Learning-based ER pipeline. The ER pipeline proceeds in four stages. First, candidate record pairs (r_i, r_j) are generated using a blocking strategy to reduce the quadratic search space. Second, a set of similarity features $\mathbf{x}_{ij}$ is computed for each candidate pair, capturing lexical and structural agreement between records. Third, a supervised classifier is trained to estimate the probability that a candidate pair refers to the same entity. Finally, predicted match probabilities are used to construct a similarity graph, which is clustered to produce the final entity partition.

Pairwise learning model. We employ ridge-regularized logistic regression as the pairwise classifier. Given a feature vector $\mathbf{x}_{ij}$ and a binary label $y_{ij} \in \{0, 1\}$ indicating whether two records match, the classifier estimates

$$p(y_{ij} = 1 \mid \mathbf{x}_{ij}) = \sigma(\boldsymbol{\beta}^\top \mathbf{x}_{ij}),$$

where $\sigma(\cdot)$ denotes the logistic function. Model parameters are learned by minimizing the regularized negative log-likelihood

$$\mathcal{L}(\boldsymbol{\beta}) = \sum_{(i,j)} \log\left(1 + \exp(-y_{ij}\boldsymbol{\beta}^\top \mathbf{x}_{ij})\right) + \lambda \|\boldsymbol{\beta}\|_2^2.$$

Ridge regularization provides numerical stability in the presence of correlated similarity features, which are common in ER tasks. Importantly, the classifier is not optimized for maximum predictive accuracy; rather, it serves as a stable and interpretable component for studying the effect of evaluation protocols.

Evaluation protocol and data dependence. A key challenge in evaluating learning-based ER systems arises from strong dependencies among records and record pairs. Because individual records participate in many candidate pairs, naive random splitting of records can lead to transductive leakage, whereby entity-specific information observed during training reappears during testing.

To address this issue, we adopt an *entity-disjoint evaluation protocol*. Ground-truth entities are first partitioned into disjoint training and testing sets,

$$\mathcal{E} = \mathcal{E}_{\text{train}} \cup \mathcal{E}_{\text{test}}, \quad \mathcal{E}_{\text{train}} \cap \mathcal{E}_{\text{test}} = \emptyset.$$

All records associated with a given entity inherit the same split. Candidate generation, feature extraction, classifier training, threshold selection, and clustering are performed independently within each split.

For comparison, we also consider a *record-level split*, in which records are randomly assigned to training and testing sets without regard to entity membership. While simple to implement, this strategy allows entities to span both splits and can inflate performance estimates. In this study, record-level splitting is included solely as a diagnostic baseline to illustrate the impact of leakage rather than as a recommended evaluation practice.

4 Experimental Setup

4.1 Datasets

We evaluate our study on two widely used benchmark datasets for entity resolution: **CORA** and **Affiliation**. The CORA dataset originates from the reference matching work of McCallum et al. (1) and is widely used as a benchmark for learning-based entity resolution. The Affiliation dataset follows the setup described in Christen (3) and has been used to evaluate ER methods under highly ambiguous, many-to-many conditions.

These datasets were selected because they exhibit contrasting structural characteristics and error modes, allowing us to examine how evaluation protocols behave under substantially different ER scenarios.

The **CORA** dataset consists of 1,879 bibliographic records describing research publications, with multiple noisy representations referring to the same underlying paper. Each record contains a rich set of structured bibliographic attributes, such as title, authors, address, publication year, and publisher-related information. CORA primarily exhibits a one-to-many duplication structure, where each real-world entity is associated with a small number of records. This makes CORA well suited for assessing the ability of learning-based ER systems to generalize pairwise matching decisions and reconstruct relatively well-defined entity clusters.

In contrast, the **Affiliation** dataset comprises 2,264 raw affiliation strings extracted from publication PDFs. Each record is an unstructured text string that may contain multiple address components, such as institution name, department, city, or country, often appearing in varying orders and formats. The dataset exhibits a many-to-many structure, in which a single affiliation string may implicitly refer to multiple entities, and the same institution may appear under numerous naming conventions. As a result, entity boundaries in Affiliation are inherently ambiguous, and over-merging is a common failure mode.

By employing both datasets, we intentionally cover two complementary ER regimes. CORA represents a relatively well-defined duplication problem, where evaluation focuses on generalization and accurate cluster reconstruction. Affiliation represents a highly ambiguous setting, where clustering metrics may disagree and evaluation protocols play a critical role in shaping conclusions. Together, these datasets allow us to isolate and analyze the impact of data splitting strategies across both moderate and extreme ER conditions, rather than tailoring conclusions to a single dataset type.

4.2 Data Splitting and Experimental Design

To examine the effect of evaluation protocols on learning-based entity resolution, we compare two data splitting strategies that are commonly encountered in practice.

Entity-level splitting. In the entity-level split, ground-truth entities are first partitioned into disjoint training and testing sets. All records associated with a given entity are assigned exclusively to one split. This strategy ensures that no entity appears in both training and testing data, thereby enforcing a strict separation between seen and unseen entities. As a result, performance under entity-level splitting reflects the ability of an ER system to generalize matching and clustering decisions to entirely new entities.

Record-level splitting. In the record-level split, records are randomly assigned to training and testing sets, without regard to their underlying entity membership. Consequently, records referring to the same real-world entity may appear in both splits. While this strategy is simple to implement and widely used, it allows entity-specific information to leak from training to testing, potentially inflating performance estimates. In this study, record-level splitting is included as a diagnostic baseline to illustrate the impact of such leakage rather than as a recommended evaluation practice.

For each dataset and splitting strategy, we repeat the complete ER pipeline—including candidate generation, feature extraction, classifier training, threshold selection, and clustering—for $k = 10$ independent random splits. All reported results are averaged across runs, with standard deviations used to quantify variability due to data partitioning.

4.3 Evaluation Metrics

Entity resolution is fundamentally a clustering problem, yet many ER systems are trained using pairwise supervision. To capture performance at both levels, we report a combination of clustering-level and pairwise-related metrics. Each metric highlights a different aspect of ER quality, and no single metric is sufficient in isolation.

Adjusted Rand Index (ARI). ARI measures the agreement between the predicted clustering and the ground-truth entity partition, adjusted for chance. It directly evaluates whether records are assigned to the correct entities and penalizes both over-merging and over-splitting. ARI is particularly informative when entity boundaries are well-defined, but it can be close to zero in highly ambiguous settings.

Normalized Mutual Information (NMI). NMI quantifies the amount of information shared between two clusterings. Unlike ARI, it is less sensitive to exact partition recovery and instead reflects how well the overall structure of the clustering aligns with ground truth. NMI can increase under aggressive merging, making it important to interpret alongside other metrics.

Variation of Information (VI). VI measures the distance between two clusterings in terms of information loss and gain. Lower values indicate greater similarity. VI provides a complementary perspective to NMI by explicitly quantifying structural divergence, and it is particularly useful for diagnosing cases where clusters become more compact but less accurate.

Pairwise F1 (PairF1). PairF1 evaluates the correctness of predicted matching pairs, balancing pairwise precision and recall. It reflects how well the system identifies individual matching decisions but does not account for transitivity or global clustering structure.

Fowlkes–Mallows Index (FMI). FMI is the geometric mean of pairwise precision and recall at the clustering level. In our experiments, FMI largely tracks PairF1 and serves as a consistency check rather than a primary diagnostic metric.

Decision threshold. We also report the similarity threshold selected during training, as it provides insight into the confidence of the learned classifier. Systematic differences in threshold values across splitting strategies can indicate the presence of leakage or memorization effects.

By jointly considering these metrics, we are able to distinguish between information-theoretic compactness, pairwise matching accuracy, and true entity reconstruction quality, which is essential for understanding the impact of evaluation protocols in ER.

5 Results

5.1 Quantitative Results

Table 1 summarizes clustering-level and pairwise-related metrics averaged over 10 runs. Higher is better for ARI, NMI, FMI, and PairF1; lower is better for VI.

5.2 Grouped Visual Comparison

Figures 1 and 2 visualize the same comparison: entity-level versus record-level splitting, shown separately for CORA and Affiliation.

Summary of the visual comparison. The grouped plots highlight a consistent pattern: record-level splitting produces clusters that appear more compact in information-theoretic terms (higher NMI, lower VI), while cluster recovery as measured by ARI and pairwise correctness (PairF1/FMI) either stagnates or degrades. This discrepancy motivates a metric-interaction interpretation rather than reliance on a single score.

Table 1: Results averaged over $k = 10$ independent runs. Standard deviations are shown in parentheses.

Clustering-level metrics				
Dataset	Split	ARI	NMI	VI
CORA	Affiliation Entity	0.0020 (0.0034)	0.6211 (0.0099)	2.6227 (0.1119)
	Record	0.0008 (0.0018)	0.7815 (0.0054)	1.5632 (0.0570)
	Entity	0.1853 (0.0767)	0.5329 (0.0574)	2.5942 (0.4060)
	Record	0.0850 (0.0186)	0.6471 (0.0070)	2.2243 (0.0510)
Pairwise-related metrics				
Dataset	Split	FMI	PairF1	Thr
CORA	Affiliation Entity	0.0114 (0.0059)	0.3299 (0.1266)	0.164 (0.0184)
	Record	0.0049 (0.0026)	0.1078 (0.0289)	0.166 (0.0190)
	Entity	0.2899 (0.0735)	0.9389 (0.0656)	0.332 (0.0839)
	Record	0.1491 (0.0206)	0.8805 (0.0333)	0.360 (0.0163)

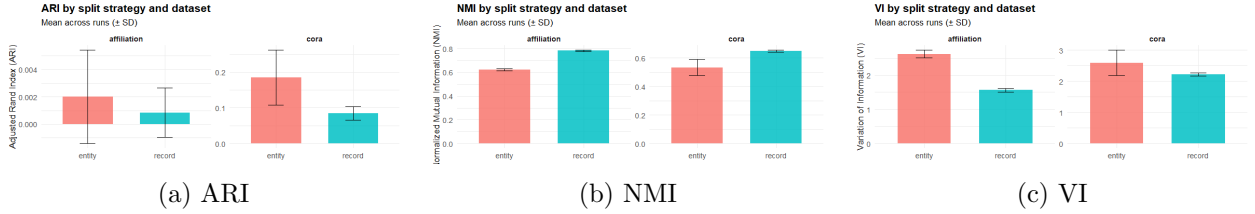


Figure 1: Clustering-level metrics under entity-level and record-level splitting. Bars indicate mean across runs; error bars show one standard deviation.



Figure 2: Pairwise-related metrics and selected decision thresholds under different splitting strategies.

6 Analysis and Discussion

CORA: generalization versus memorization. For CORA, pairwise matching is comparatively easy (Table 1): PairF1 is high under both protocols (0.94 entity-level; 0.88 record-level). Nonetheless, entity-level splitting achieves substantially higher ARI (0.19 vs 0.085), indicating better recovery of entity partitions. Record-level splitting again increases NMI and decreases VI, but these gains coincide with reduced ARI and reduced FMI, consistent with leakage-driven tightening of clusters rather than improved entity reconstruction. The larger ARI variance under entity-level splitting reflects genuine sensitivity to unseen-entity generalization, which is expected in a realistic evaluation.

Affiliation: ambiguity and metric divergence. For Affiliation, ARI remains near zero under both splitting schemes, consistent with the dataset’s many-to-many structure and intrinsic ambiguity. However, record-level splitting yields substantially higher NMI (0.78 vs 0.62) and lower VI (1.56 vs 2.62), suggesting more “informationally aligned” clusterings. This apparent improvement is not supported by pairwise correctness: PairF1 drops sharply under record-level splitting (0.11 vs 0.33), with FMI similarly small in both settings. Taken together, these results indicate that record-level splitting encourages over-merging that reduces entropy (improving NMI/VI) while increasing false matches, a failure mode that would be missed if evaluation relied solely on information-theoretic metrics.

Role of VI and threshold behavior. Including VI clarifies an important ambiguity: lower VI under record-level splitting does not necessarily signal improved ER, because VI can decrease under aggressive merging. The threshold statistics provide an additional cue. Record-level splitting selects slightly higher thresholds (notably for CORA), consistent with more confident clustering when entity-specific patterns repeat across train/test. Under entity-disjoint splits, threshold variability increases, reflecting uncertainty when generalizing to unseen entities rather than instability of the pipeline.

Overall, the results support a conservative conclusion: splitting strategy can dominate observed performance, and entity-disjoint evaluation is better aligned with the goal of assessing generalization to unseen entities.

7 Limitations

This study is evaluation-centric and does not propose a new ER model. Accordingly, conclusions are drawn using a single stable classifier; more expressive models (e.g., boosted trees or neural matching) may exhibit different quantitative sensitivity to splitting, even if the qualitative leakage mechanism remains.

We report results on two benchmark datasets with distinct characteristics, but these do not cover all ER settings, such as temporal drift, cross-domain matching, or highly heterogeneous

schemas. Extending the evaluation to additional datasets is a natural next step.

Finally, although ARI, NMI, VI, FMI, and PairF1 capture complementary aspects of ER quality, no single metric fully characterizes downstream utility. Future work could connect evaluation protocols to application-specific objectives (e.g., decision-making errors induced by over-merging versus under-linking).

8 Conclusion

We examined the effect of data splitting strategies on learning-based entity resolution. Record-level splitting can introduce transductive leakage and systematically alter metric conclusions, often yielding more compact clusterings that are not necessarily more accurate. Entity-disjoint evaluation provides a principled alternative that better reflects generalization to unseen entities and supports more reliable method comparison.

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